

Verifiable AI Explanations for Orbital Capacity and Space Sustainability Analysis

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ABSTRACT

The decision value of orbital-capacity and space-sustainability models depends not only on numerical outputs, but on whether artificial intelligence explanations preserve the evidence, assumptions, provenance, and limits governing those outputs. This work presents Catalyst, an integrated simulation-grounded explainable artificial intelligence (XAI) system for Space Domain Awareness (SDA) and Space Situational Awareness (SSA), orbital-capacity analysis and sustainability decision support, in which E-MOCAT artifacts and registered technical, policy, and economic sources retain evidentiary authority while language models generate bounded explanations.

The primary matched experiment compares a Reference Code World Model (CWM) with a Catalyst-Adapted CWM within the same Catalyst pipeline across 800 fixed cases in eight equally weighted sections of 100. Prompts, model-visible evidence, case order and decoding are held fixed across both configurations; the protected references are likewise fixed, and are withheld from both models. A response is successful when it provides the form of answer justified by the evidence (direct, qualified, or an evidence-bound non-answer) and remains grounded, internally consistent, relevant, appropriately scoped, and free of material error.

Successful responses increased from 660/800 (82.50%) to 785/800 (98.13%), a gain of 125 responses and 15.63 percentage points, with improvement in all eight sections. A separate 120-case, six-configuration study compared Reference and Catalyst-Adapted configurations within each of CWM, Gemma, and Nemotron; its structured results showed model-family dependence rather than a universal adaptation effect. Paired contrasts, causal-scope cases, evidence ablations and conflicts, repeatability probes, and irrelevant-metadata controls provided contrastive tests on structured response fields. Catalyst therefore operationalizes XAI principles as system-design constraints and makes the resulting explanation behavior measurable. Because those constraints are identical in both configurations, the matched comparison isolates what Catalyst-specific adaptation contributes within that architecture.

1 The Verification Gap

Low Earth orbit (LEO) capacity cannot be read from a single population count [1]. Population and shell density, conjunction and collision behavior, and fragmentation evolve together [2–4]. Launch activity and mitigation assumptions further shape that evolution [5–7], while object-resolved Monte Carlo and shell-level source–sink models make those relationships quantitative [8–10]. Their outputs stay conditional on catalog epoch, scenario configuration, metric definition, horizon and stochastic realization [11], so an explanation is decision-useful only while those conditions remain attached to it.

A language model may state a plausible trend while losing the numerical basis, causal driver, provenance, uncertainty, or scope that makes it supportable. Prior work separates persuasive language from explanation that is attributable to an identified source [12–15]; space-operations work has investigated explainable natural-language methods for satellite scheduling [16]; and trustworthy-AI guidance ties explainability to transparency, documentation, and evaluation [17]. The practical question is therefore not whether a model can narrate a result, but whether that narration can be tested against the record it came from.

Catalyst addresses that question by preserving an inspectable chain from the task and model-visible evidence to the selected model configuration, generated explanation, and successful response judgment (Fig. 1). E-MOCAT artifacts and registered sources retain authority for validated quantities, assumptions, and source claims; the language model, here a Code World Model (CWM) [18], performs interpretation and synthesis, while Catalyst evaluates whether the visible response remains supportable. The model family, the adaptation state, the task formulation, and the evidence condition can each be varied under controlled conditions, enabling contrastive XAI.

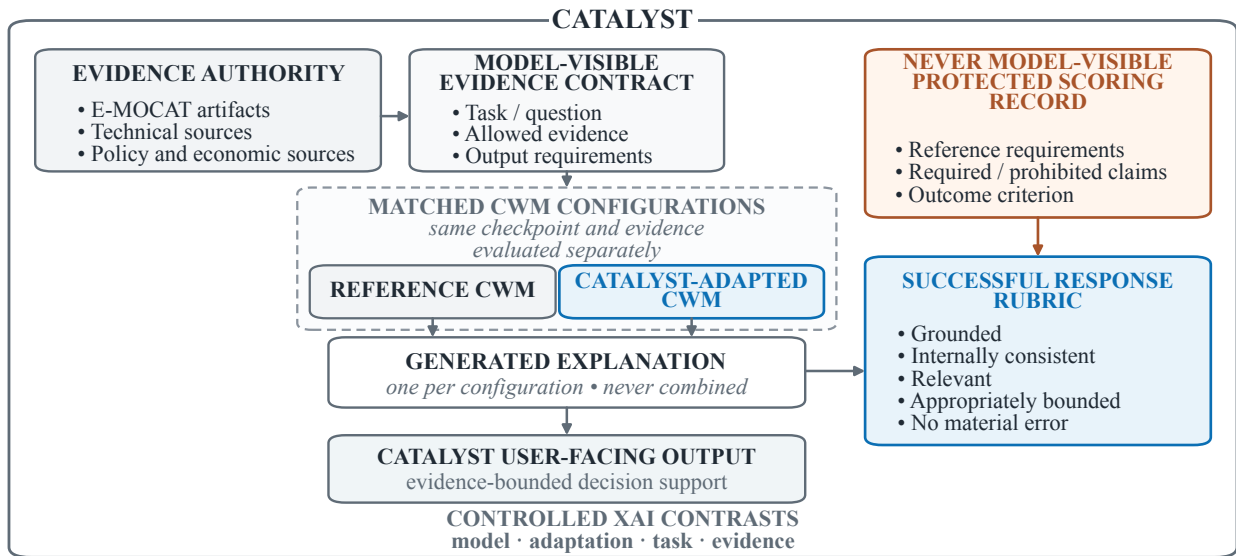


Fig. 1. Catalyst XAI architecture and matched CWM comparison. E-MOCAT artifacts and registered sources retain evidentiary authority. Both CWM configurations receive identical model-visible evidence; the protected reference and the Successful Response rubric remain withheld. Only visible responses are assessed.

Catalyst applies explainable AI to Space Domain Awareness (SDA) and Space Situational Awareness (SSA) tasks: orbital-environment interpretation, conjunction and collision reasoning, provenance, policy, economics and evidence-bounded decision support. The evidence comes from a simulation-driven orbital-debris governance application and spans physical, policy and economic material [19–22] that an explanation must keep intact. The unit of assurance is a preserved relationship among the question, model-visible evidence, model configuration, generated response and explicit criterion, rather than a hidden reasoning trace.

2 Catalyst: Integrated, Authority-Preserving XAI

Catalyst is the complete software system connecting E-MOCAT simulation artifacts (object-resolved Monte Carlo and shell-level source–sink outputs with their scenario settings, seeds and time histories), registered technical, policy and economic sources, language-model configurations, and user-facing outputs. The language model is one system component, and the Catalyst-specific adapter is one configuration of that component. E-MOCAT artifacts and registered sources retain authority for validated quantities, assumptions and source claims; the model interprets, and Catalyst preserves the evidence needed to assess whether that interpretation is supportable.

Each case contains two separate records: a model-visible evidence contract carrying the task, permitted artifacts and output requirements, and a protected scoring record carrying the reference requirements and outcome criterion, withheld from both configurations. Together they make it possible to reconstruct what the model was asked, what it could use, what it said and why it succeeded or failed.

2.1 Successful Responses as an XAI Endpoint

A response is successful when it provides the form of answer justified by the available evidence: a direct evidence-supported answer when the record is sufficient, a qualified or partial interpretation when support is limited, or an explicit evidence-bound non-answer when the conclusion cannot be justified [23]. These are alternative valid response forms, not success tiers. Every successful response must remain evidence-grounded, internally consistent, relevant, appropriately bounded, and free of material error. An unsupported refusal on an answerable case does not succeed.

The endpoint measures observable response supportability, not exact wording or hidden reasoning. Numerical fidelity, causal validity, provenance and epistemic scope are assessed within that composite rubric and are not reported as separate rates; semantic stability and contrastive sensitivity are assessed separately by the probes described in Section 5.

3 A Matched 800-Case Evaluation

The primary evaluation contains 800 fixed cases divided equally among eight sections spanning backend interpretation, policy, economics, orbital reasoning, E-MOCAT population analysis, tool execution, evidence and provenance, and cross-domain synthesis, each holding 25 foundational, 25 intermediate, 25 difficult and 25 adversarial cases and 70 answerable, 15 partial-evidence and 15 insufficient-evidence cases (560/120/120 overall), so no one capability dominates the aggregate.

Both configurations operate within Catalyst and use the same public facebook/cwm-sft checkpoint [18]. The Reference CWM uses that checkpoint without the Catalyst-specific domain adapter loaded; the Catalyst-Adapted CWM uses the same checkpoint with the final Catalyst-specific low-rank adapter [24] enabled. The adapter was trained by supervised fine-tuning on 1,149 training cases with a further 188 development cases, 1,337 in all. These span twenty task families covering backend interpretation, orbital reasoning, policy and economics, governance, tool planning, uncertainty, evidence-limited answers and adversarial refusal, each under a fixed output contract, so that training targets the Section 2.1 response behavior rather than any benchmark answer. Training and evaluation are disjoint: across those 1,337 cases and these 800 there is no shared case identifier, no exact prompt match, no evaluation prompt sharing over half its 9-grams with a training prompt, and no protected reference appearing verbatim in training. Case identifiers, prompts, model-visible evidence, case order, and decoding are identical. The protected references are fixed across both configurations and are not visible to either model. Greedy decoding uses temperature 0.0, top- p 1.0, a 1,536-token maximum, and no inference fallback.

Every response was evaluated under the predefined binary Successful Response rubric, applied identically across both configurations and all 800 cases against the same five preserved records: the prompt, the model-visible evidence, the reference requirement, the response and its generation status. The 560/120/120 split describes how the benchmark was authored; what the visible evidence actually supports is a separate question, and it sets how each case is scored. A case is scored against the full reference conclusion (276) when the visible evidence supports it, against a partial conclusion (24) when it supports only part, and on respecting the evidence boundary for the remaining 500 (284 answerable and 96 partial cases whose conclusion exceeds what the model could see, plus all 120 insufficient-evidence cases). Material errors include unsupported numerical values, incorrect units or comparisons, contradictions of the record, fabricated sources or tool results, unsupported causal or policy conclusions, and failure to respect an evidence limitation that materially changes the answer. The outcome and its written rationale are retained for all 800 pairs.

For each configuration, the Successful Response Rate is $\hat{p} = x/800$, with Wilson 95% intervals [25], and an exact two-sided McNemar test [26] evaluates the paired asymmetry across the discordant cases.

A separate 120-case experiment produced 720 responses from Reference and Catalyst-Adapted configurations within each of CWM, Gemma [27] and Nemotron [28]. Within family pairs used the same frozen cases and evidence; cross-family differences are descriptive, since architectures, chat templates and runtime

versions differ. The companion X0 test suite, comprising paired contrasts, causal-scope cases, evidence ablations and conflicts, repeatability probes and irrelevant-metadata controls, examines how explanation behavior changes across model families, adaptation states and evidence conditions. Its denominator and endpoint stay separate from the Successful Response Rate: it scores structured response fields, not prose.

4 Successful Responses Across Every Section

Within Catalyst, Catalyst-specific CWM adaptation increased successful responses from 660/800, or 82.50% (Wilson 95%: 79.71–84.98%), to 785/800, or 98.13% (96.93–98.86%): a gain of 125 responses and 15.63 percentage points. Of the matched pairs, 652 were successful under both configurations, 133 under the Catalyst-Adapted configuration only, 8 under the Reference configuration only and 7 under neither; the exact McNemar result on the 141 discordant pairs, $p = 2.414 \times 10^{-30}$, indicates a strong configuration-level difference (Fig. 2).

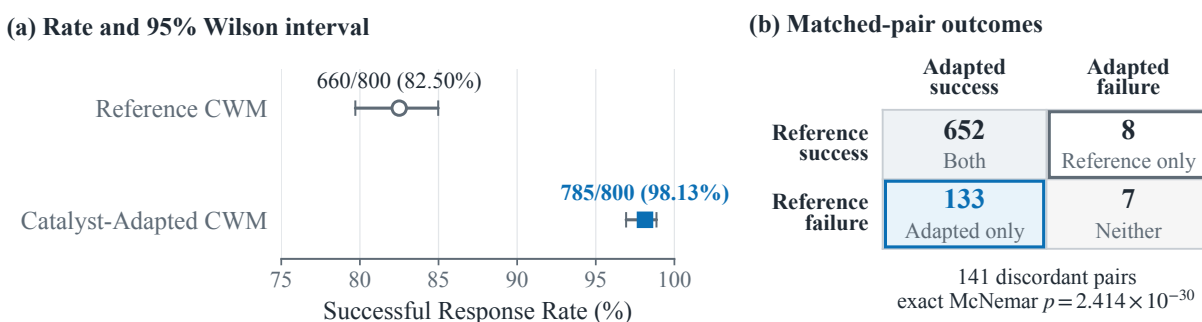


Fig. 2. Overall matched Successful Response Rate. Reference CWM 660/800 (82.50%); Catalyst-Adapted CWM 785/800 (98.13%), with Wilson 95% intervals. The paired outcomes support the exact McNemar comparison.

Performance improved in all eight sections (Fig. 3a): the Catalyst-Adapted CWM reached 91–100 successful responses per section, with seven at 98–100. The largest gains were orbital/conjunction/collision reasoning (+30 percentage points) and evidence/provenance/safety (+25); the rest ranged from +3 to +18, tool/scenario execution lowest at 91/100. The improvement spans physical interpretation, policy and economic reasoning, provenance and cross-domain synthesis rather than one capability.

5 Comparative XAI for SDA, SSA and Space Sustainability

The constraints that make explanation behavior observable (authoritative artifacts, a bounded evidence contract, preserved provenance, and an answer form matched to the record) hold in both configurations. The matched comparison therefore isolates what Catalyst-specific adaptation adds inside that architecture, raising successful responses from 82.50% to 98.13% across technical interpretation, evidence handling and cross-domain synthesis alike, as required for SDA and SSA applications.

The companion X0 tests add contrastive XAI at the structured-claim level. Responses were scored on seven content dimensions: verdict, numeric value and unit, direction, causal factor, uncertainty, provenance, and declared limitations; each was also scored on whether it emitted a *response-contract object*, defined as a complete JSON object meeting the schema. CWM numerical-and-unit correctness rose from 89/120 to 114/120 while Gemma fell from 105/120 to 93/120. Nemotron completed more generations under adaptation (truncations fell from 72/120 to 47/120), but that did not carry through to conformance: well-formed, complete response-contract objects moved only from 48/120 to 49/120.

The X0 tests provide contrastive evidence by identifying which structured behaviors remain stable and which depend on model family or adaptation. Two outcomes held across every configuration: no fabricated artifact identifier appeared in any of the 720 responses, and responses to each configuration’s five

repeated cases were reproduced byte for byte. Irrelevant-metadata invariance, on five perturbed cases per configuration, held 4/5 for Catalyst-Adapted CWM and Catalyst-Adapted Gemma but 1/5 for Nemotron in either configuration. Unsupported-claim avoidance on the 21 knowledge-limit cases (counting a qualified answer that withholds the unsupported claim, not only an outright non-answer) rose with adaptation in every family: 0 to 5 for CWM, 0 to 2 for Gemma, and 3 to 6 for Nemotron. The strictest probes require every scored dimension to be correct at once and were satisfied at most 1/30 on paired contrast, 1/20 on causal scope, and 2/15 on evidence ablation; these counts measure a harder property than the primary endpoint and are reported separately. Read together, the X0 tests demonstrate exact repeatability across all tested repeated cases, complete artifact-identifier discipline across all 720 responses, and improved unsupported-claim avoidance in every adapted family. The tests connect controlled changes in evidence condition to observable changes in structured output, providing the intended contrastive evidence.

When grouped by the authored strata rather than the Section 3 scoring bases, adapted success reached 97.68% on answerable and 99.17% on both partial and insufficient evidence cases; the model held the evidentiary boundary rather than inventing one.

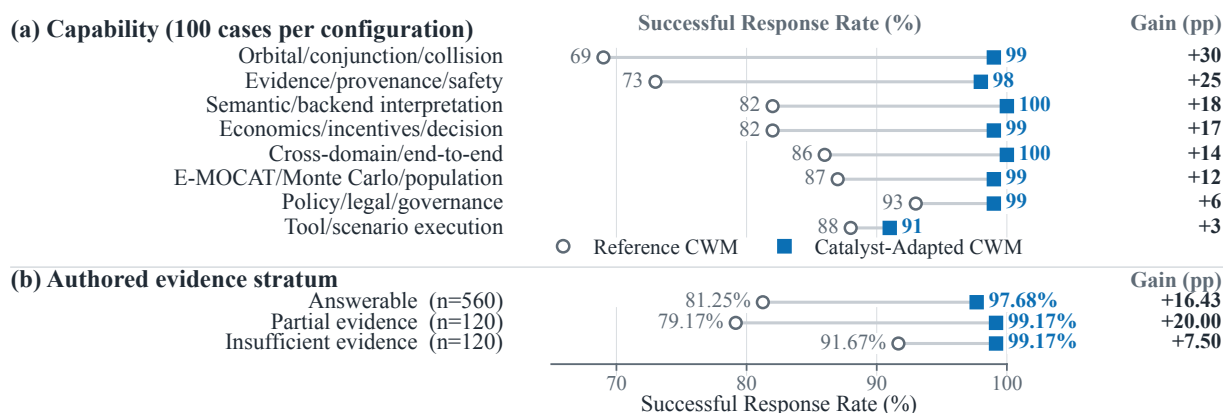


Fig. 3. Successful responses by capability and by authored evidence stratum. All eight sections improved. The lower panel keeps the original 560/120/120 benchmark groupings, distinct from the model-visible scoring bases of Section 3.

6 Limitations and Scope

Three boundaries govern these results. The Successful Response Rate is this study's endpoint, a predefined binary supportability measure at the visible interface scored against the preserved record; it is not exact-match accuracy or internal reasoning fidelity and does not validate the physical E-MOCAT artifacts, which keep their own implementation, scenario and validation limits. The primary statistic isolates one CWM checkpoint and one Catalyst-specific adapter; it does not isolate the surrounding XAI constraints, which were fixed across both configurations and so enable the measurement rather than being measured by it. The six-configuration study uses a different denominator and scores structured fields, not prose, and neither experiment establishes universal model transfer or reliability under fully autonomous operation.

7 Conclusion

Catalyst operationalizes XAI by measuring explanation behavior under fixed evidence rather than relying on post hoc descriptions. Within this architecture, adaptation raised successful responses from 660/800 (82.50%) to 785/800 (98.13%), a gain of 125 responses and 15.63 percentage points, with all eight sections reaching 91–100. Together with X0 evidence on stability and evidence sensitivity, these results support responsible AI assistance for SDA, SSA and space sustainability while keeping scientific authority with the evidence.

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